Pattern Recognition.

Linear algebra
probability
statistics } self study

probability Density function. (PDF)

over n- dimensional Euclidean space

$x \in \mathbb{R}^n$

$\vec{x} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

$p(\underline{x}) \geq 0 \quad \forall \; x \in \mathbb{R}^n$

$\int_{\mathbb{R}^n} p(\underline{x}) \, d\underline{x} = 1$

$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

$$p(\underline{x}) = \frac{1}{(\sqrt{2\pi})^n |\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(x-\underline{\mu})' \Sigma^{-1} (x-\underline{\mu})\right\}$$

$1 \times n \qquad n \times n \qquad n \times 1$

$\Sigma \geq 0$
↳ covariance.

- less than zero - complex

$x_1, x_2, \dots, x_n \in \mathbb{R}$

$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i \longrightarrow$ mean of nth observation

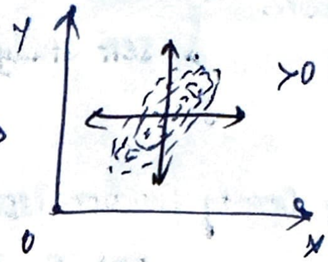
Varience.

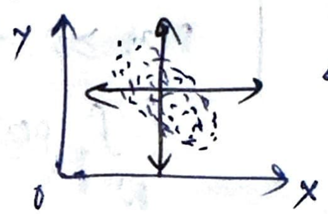
$var(x) = 1/n \sum_{i=1}^{n} (x_i - \bar{x})^2$

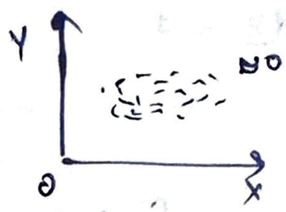
↳ $1/_{n-1} \sum_{i=1} (x_i - \bar{x})^2$

height $\uparrow$ $\rightarrow$ weight
(X, Y)

Covariance : $\boxed{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}$







if the 'n' features are $X_1, X_2, \ldots, X_n$
then VARIANCE - COVARIANCE matrix is of these 'n' features
denoted as Σ is defined by

$$\Sigma = \begin{pmatrix} cov(x_1, x_1) & cov(x_1, x_2) & \cdots & cov(x_1, x_n) \\ cov(x_2, x_1) & cov(x_2, x_2) & \cdots & cov(x_2, x_n) \\ \vdots & \vdots & & \vdots \\ cov(x_n, x_1) & cov(x_n, x_1) & \cdots & cov(x_n, x_n) \end{pmatrix}$$

$X \longleftrightarrow Y$

$d(x, y) \geq 0$

$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

eg: $\begin{pmatrix} 160\,cm \\ 70kg \end{pmatrix}$ $\begin{pmatrix} 150cm \\ 72kg \end{pmatrix}$
X Y

eg: $d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i-y_i)^2}$ $\quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

$= \sqrt{13}$

Irrespective of the units, distance should be same

unit change - distance should not change

$$d^2(x,y) = (x_1-y_1)(x_2-y_2)\begin{pmatrix}1&0\\0&1\end{pmatrix}\begin{pmatrix}x_1-y_1\\x_2-y_2\end{pmatrix} \geq 0$$

$$\underbrace{(x_1-y_1)(x_2-y_2)\begin{pmatrix}w_1&0\\0&w_2\end{pmatrix}\begin{pmatrix}x_1-y_1\\x_2-y_2\end{pmatrix}}$$

$\begin{pmatrix}w_{11}&w_{12}\\w_{21}&w_{22}\end{pmatrix}$ ↳ generic matrix.

for any value of $(w_1, w_2) \geq 0$

Symmetric matrix

if $(w_1, w_2) > 0$

↳ strictly greater than 0 → it is greater than 0.

* two persons feature → identity
distance = zero.

* if distance ~~similar~~ are equal → two features are similar

$$\Rightarrow \underbrace{(x_1-y_1)\,(x_2-y_2)}_{a'}\underbrace{\begin{pmatrix}2&-1\\-1&2\end{pmatrix}}_{A}\underbrace{\begin{pmatrix}x_1-y_1\\x_2-y_2\end{pmatrix}}_{a} > 0 \text{ if } x_1 \neq y_1 \text{ (or) } x_2 \neq y_2$$

* $A_{n\times n}$ is said to be positive definite if $a'Aa > 0$ $\forall a \neq \begin{pmatrix}0\\0\\ \vdots\\0\end{pmatrix}_{n\times 1}$

$$\underset{1\times n}{a'}\ \underset{n\times n}{A}\ \underset{n\times 1}{a}$$

* $A_{n\times n}$ is said to be positive semi definite if $a'Aa \geq 0$ $\forall a$.

* VARIANCE - COVARIANCE matrix is non-negative definite.

if Σ is positive definite, the some perspectives follow automatically

- Determinant of a matrix if product of the eigen values if it is Σ then eigen values ≥ 0, for non-negative definite
  eigen values > 0, for +ve definite

31st July'25

A metric space is a set together with a notion of distance b/w its elements, called points the distance is measured by a function called a metric / distance function.

A metric space is an ordered pair (M, d) where M is a set and d is a metric on M i.e a function.

$$d : M \times M \to \mathbb{R}$$

satisfying the following:

$x, y, z \in M$

1) $d(x, x) = 0$
2) $d(x, y) > 0$, if $x \neq y$
3) $d(x, y) = d(y, x)$ (symmetry) (distance is similar from both ends

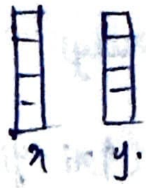
Triangle inequality:

$$\boxed{d(x,y) + d(y,z) \geq d(x,z)}$$

$\sqrt{(x_1-y_1)^2 + (x_2-y_2)^2} \rightarrow$ distance function

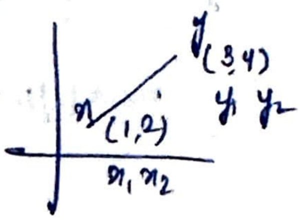
$(x_1-y_1)^2 + (x_2-y_2)^2 \rightarrow$ distance function?

Calculating distance between two vectors:



Euclidean distance.

$$d(x,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$



eg: $\rightarrow \sqrt{(1-3)^2 + (2-4)^2}$

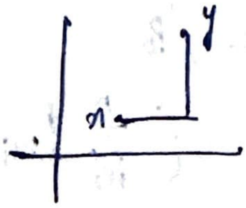
$\Rightarrow \sqrt{8} = 2.82$

Manhattan / City Block / Taxi cab distance. (absolute distance)

$$d(x,y) = \sum_{i=1}^{n} |x_i - y_i|$$

eg: $\rightarrow |1-3| + |2-4|$

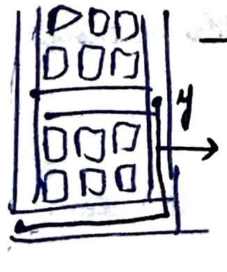
$\rightarrow 4$



$x: \{x_1, x_2, x_3 \dots x_n\}$

$y: \{y_1, y_2, y_3 \dots y_n\}$

↳ Applications:



→ city of houses.

→ reaching x to y through Manhattan distance

② occlusion

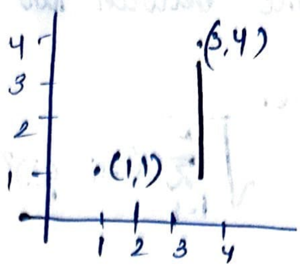
A NORM is a function from a real / complex vector space to the non negative real no's that behaves in certain ways like the

Chebyshev distance (self study)

$$d(x,y) = \max_i (|x_i - y_i|)$$

$$\max(|1-3|, |1-4|)$$

$$= \max(2,3) = 3$$



metric

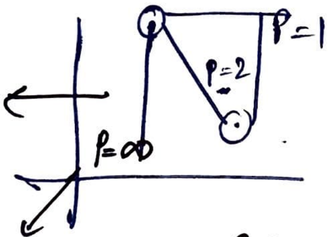
L_∞

→ The chebyshev distance is the limiting case of the order-p Minkowski, when p reaches infinity.

Minkowski distance : L_p

$$d(x,y) := \left(\sum_{i=1}^{n} |x_i - y_i|^p\right)^{1/p}$$

Chebyshev distance



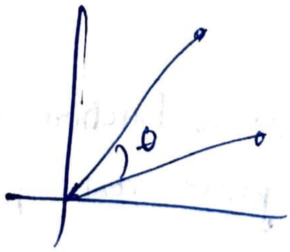
hyper parameter (change with dataset)

Cosine similarity: similarity b/w two non zero vectors

[-1, 1] → vectors.

opposite direction

same orientation



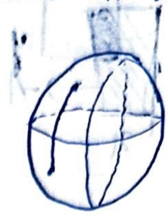
$$\sqrt{\sum_{i=1}^{n} x_i^2}$$

$$\underbrace{x \cdot y}_{\text{non zero vectors}} = \|x\| \, \|y\| \cos\theta$$

$$d(x,y) = \cos\theta = \frac{x \cdot y}{\|x\| \, \|y\|}$$

Havereine distance (self study) - distance between 2 points on a sphere

formula:



Hamming distance. ideally - 2.

0 0 1 1 0 1
0 0 1 0 0 0
↑ 2 ↑

2/6 → total digits.
↓
differences

0 0 0 0
1 1 1 1 } 4 / 4/4 = 1 → total no. of digits.
↓
differences for both numbers.

eg: Movie | action | comedy | romance.

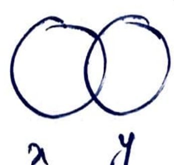
ground truth

predicted truth. | action | horror | romance. } hamming distance = 1.

predicted truth | horror | action | romance → hamming distance - 2

Jaccard index / intersection over union (not a distance)

$$IOU = \frac{|x \cap y|}{|x \cup y|} - J(x,y)$$



$$0 \leq J(x,y) \leq 1$$

$|x \cap y| = 0$

$|x \cap y| = |x \cup y| = x = y$

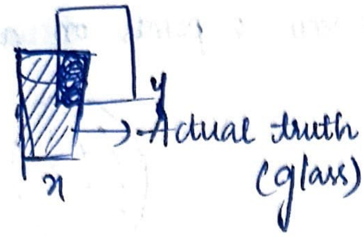
Jaccard distance → overlapping - distance (0) so, 1

$$d_J(x,y) = 1 - J(x,y)$$

$$= \frac{|x \cup y| - |x \cap y|}{|x \cup y|} = \frac{x \Delta y}{|x \cup y|}$$

$x \Delta y$ → symmetric difference over $|x \cup y|$ → union

Use case :



$\frac{x \cap y}{x \cup y} \geq 0.5$

performance metrics - 0.5 (threshold)
0.55
0.6
so, on.

→ $IOU_{0.5}$ $IOU_{0.6}$ IOU_{7} IOU_{8} IOU_{9}

$IOU_{\mu} = IOU + IOU + IOU + IOU + IOU/\mu$

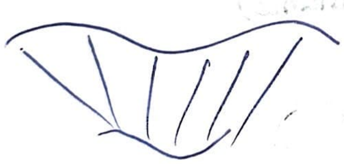
$= 0.5+0.6+7+8+9/\mu$

Sorensen Dice index.
DTW (Dynamic Time working) – series/ continuous data.

} self study

measures the distance b/w 2 time series of diff length

↓
spend time taking data



Mahalanobis distance (MD)

It is a measure of the distance b/w a point (x) & a probabilistic distribution Q.

unknown

probability distribution Q on $\mathbb{R}^n$ with mean $\mu = (\mu_1, \mu_2, \ldots \mu_N)^T$

MD of a point x = $\cancel{\sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}}$ $(x_1, x_2, \ldots x_n)^T$ from Q is

$d_M(x, Q) = \sqrt{(x-\mu)^T \Sigma^{-1} (x-\mu)}$ → gaussian distribution

($\because \Sigma$ is positive semi definite

so, is Σ^{-1} (thus the square roots are always defined)

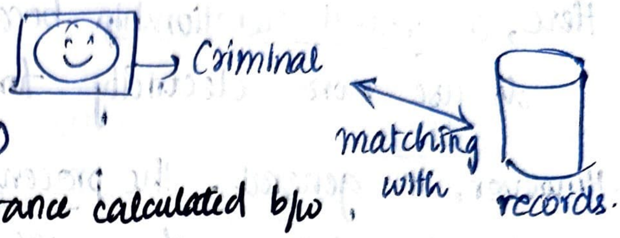
Bhattacharya distance. (self study)

<u>4th Aug'25</u>

pattern matching:

↳ check distance function.

if $D(x,y) \approx 0$ (decide some threshold)



→ mahalanobis distance function distance calculated b/w (one point to a distribution)

Σ^{-1}

→ chaeck Σ is non-negative distance.

→ check mean, covariance, variance., <u>correlation.</u>

CORRELATION / DEPENDENCE

is any statistical relationship, whether <u>casual</u> (↗ cause) or not b/w '2' random variables or bivariate data

correlation may indicate any type of association but in statistics, it usual refer to the degree to which a pair of variables are linearly relate

eg: Dependent phenomena include the correlation

1) b/w the height of parents & their offspring

2) correlation b/w price of goods / product & quantity of the consumers are willing to purchase as it is depicted in <u>Demand curve</u> (self study)

CORRELATIONS are useful, since ~~they are~~ can indicate a predictive relationship that can be exploited in practise. (useful for prediction purpose)

eg: a electrical utility may produce less power on a mild day based on the correlation b/w electricity demand and weather.

Here, a causal relationship because of extreme weather causes people to use more electricity for cooling

However, in general, the precense of correlation is not sufficient to infer the presence of causal relationship.

CORRELATION does not imply CAUSATION

In logic, the technical use of word implies. means sufficient condition.

$\boxed{p \to q}$ p implies q
if p then q,
if p is true, then q follows.

cause can refer to necessary sufficient or contributing causes.

eg: the faster that windmills are observed to rotate, the more wind is observed. Therefore wind is caused by the rotation of windmills.

here, correlation (simultaneously) b/w windmill activity and wind velocity does not imply that wind is caused by windmills.

it is rather the other way around, as suggested by the fact that that wind does not need windmills to exist while windmills need wind to rotate

wind can be observed in places, where there are no windmills as well (wind existed before the windmills are invented)

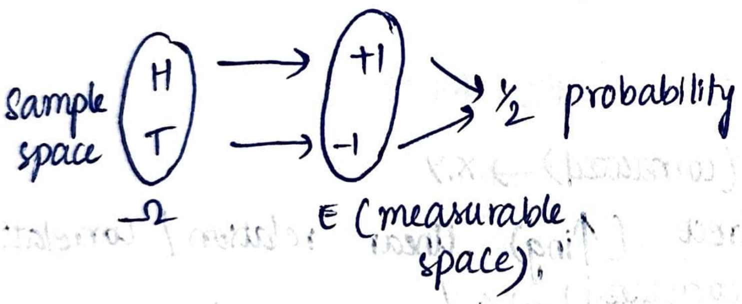
Random variable / Random quantity / ALEATORY or STOCHASTIC variable.
↳ function

Is a mathemactical formalization of a quantity which depends on random events.

the term 'Random variable' in its mathematical definition refers to neither randomness nor variability, but instead of a math function in which.

→ the domain is a set of possible outcomes in a sample space.
(eg: Head / Tail from coin flipping. {H, T}
↳ event

→ the sample is a measurable space.
eg: {-1, 1}, if H maps to -1
T maps to 1

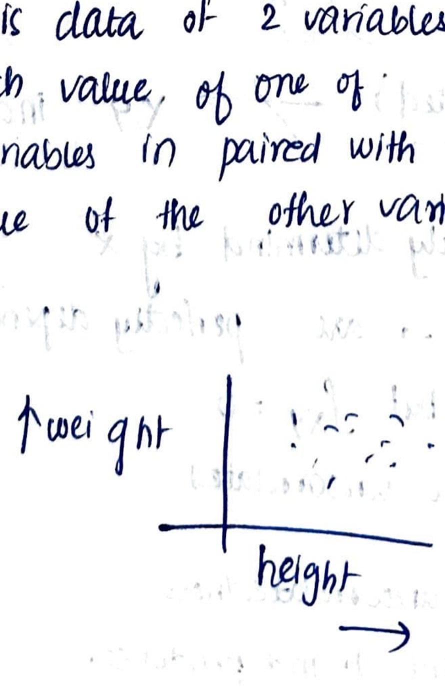


A random variable X is a measurable space. Ω in space E

Bivariable data is data of 2 variables, where each value of one of the variables in paired with a value of the other variable.

multivariable data (self study)

Scatter plot



PEARSON CORRELATION coefficient: ρ_{XY}
(PCC)

measures linear correlation b/w 2 sets of data.

$$\boxed{\underline{\rho_{XY}} = \frac{cov(X,Y)}{\sigma_x \sigma_y}}$$

$\downarrow$

normalized measurement of covariance

σ_X - standard deviation of x

σ_Y - standard deviation of y

such that $\boxed{-1 \leq \rho_{XY} \leq 1}$

special case $-1, 0, 1$

z- normalization
min-max normalization. } (self study)

objective of normalization -
(mapping to certain range)

Correlation and Independence

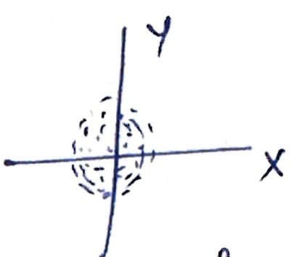
$\rho_{XY} \rightarrow$ is 1 for perfect direct ($\uparrow$ing) linear relation / correlation (correlated) $\rightarrow$ X,Y

$\rho_{XY} \rightarrow -1$ for perfect inverse ($\downarrow$ing) linear relation / correlation. (anti correlated) $\rightarrow$ X,Y

XY independent $\rightarrow$ $\rho_{XY} = 0$ (X, Y, uncorrelated)
if true guaranteed and not viceversa.

$\downarrow$

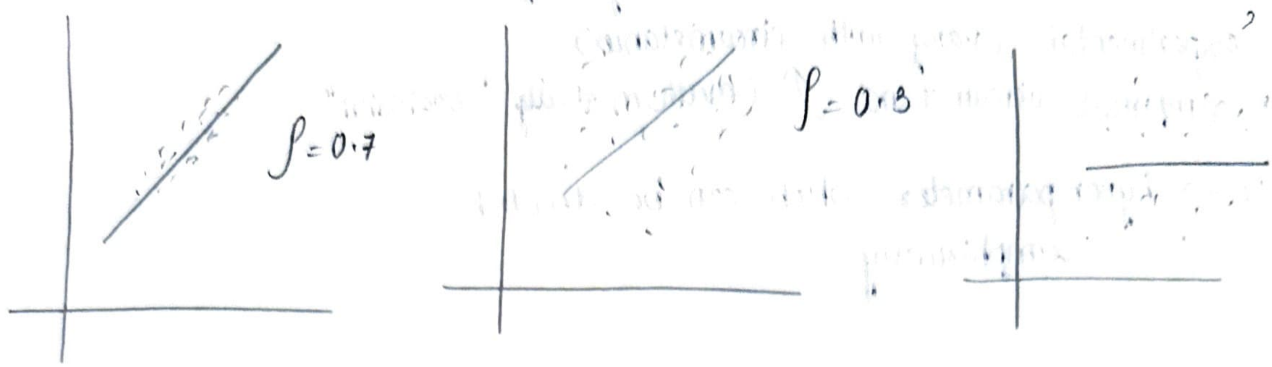
$\rho_{XY} = 0$ (X,Y uncorrelated) $\not\rightarrow$ X,Y independent



$Y = X^2$

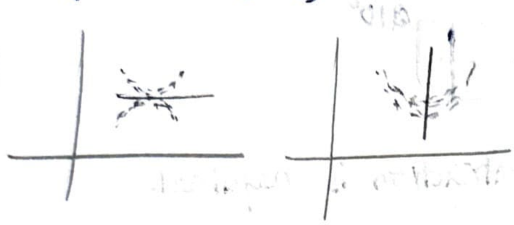
Y is completely determind by X
X,Y $\rightarrow$ are perfectly dependent
but $\rho_{XY} = 0$
they are uncorrelated

where X,Y are jointly normal, uncorrelatedness is equivalent to independence.



* more the value near to zero the more they are scatted
* symmetry (then $\rho = 0$)

Eg:



SPEARMAN's RANK correlation coefficient.

→ problem definition - computation problem. & perform ~~our~~ pattern matching. Extraclass - 4/8/25

social problem converted to ↗

so, differentiate it into inputs & outputs
↳ what clients wants

↓ given by client (or) synthetically generate

B 8 → 2 characters

↘ ↙ matching → exactly point by point → pixels 100 (matching 100 times)

↙ ↘ Yes NO

$\boxed{B}$ 10X10 $\boxed{8}$ 10X10

converting to array

$\begin{bmatrix} a_1 \cdots a_{10} \\ \quad a_{100} \end{bmatrix}$ 10X10 $\begin{bmatrix} b_1 \cdots \\ \quad b_{100} \end{bmatrix}$ 10X10

flatten (process)

a_1	a_2	......	a_{100}

vector

b_1	b_2	......	b_{100}

2) finding distance (using distance functions)

$d(A,B) >$ some threshold → not matching

$0 \le d(A,B) \le$ Threshold → matching

how to choose Threshold? can have many ways.
↳ experimental (vary with circumstances)
↳ empirical circumstance ↑ (mathematically "constraint"

T → hyper parameter which can be decided emphirically.

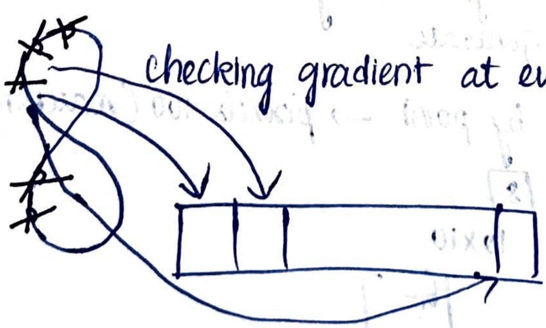
→ if $\boxed{B}_{1000 \times 1000}$ $\boxed{8}_{1000 \times 1000}$

$\boxed{a_1 \mid a_2 \mid \cdots \mid a_{100}}^{10^6}$ $\boxed{a_1 \mid a_2 \mid \cdots \mid \cdot}^{a10^6}$

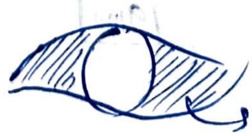
matching is difficult - good feature extraction is required.

vertical symmetry
distance between two points } solutions. in B it is more  (similar distances encountered)
and in 8 it is less.  (all different distances encountered in this case)

Hand engineered feature extraction
Hand crafted feature extraction


checking gradient at every point (gradient chain code)

(self study)
Local Binary pattern ~~also~~ ~~[illegible]~~ ~~as~~ Textural feature.
LBP.


sihlera pattern matching

Q)

intersection points / colour inversion.

finding hole → algorithm.

Q)

1X30

| .5 | .1 | .3 | ... | .3 |

P_1

stock for last 30 days

| .2 | .1 | .3 | ... | .5 |

P_2 1X30

stock → same/not

$$\left.\begin{array}{l} D(P_1, P_2) > T \rightarrow \text{different} \\ D(P_1, P_2) \leq T \rightarrow \text{same.} \end{array}\right\} \text{smaller data}$$

if for larger data (making some intervals & average it down - solution)

Gene Sequence Matching:

P_1 | A | T | G | C | ---- A G G C ...

P_2 | | T | A | G | C ---- T T A G ....

$$\left.\begin{array}{l} A \rightarrow .25 \\ T \rightarrow .50 \\ G \rightarrow .75 \\ C \rightarrow 1 \end{array}\right\} \text{find distance}$$

Training data known sample.

6th Aug'25

topics covered till now → Use cases
Patten Matching
↳ Distance function
↳ Mahalanobis distance
↳ Variance - Covariance Matrix Σ, Σ^{-1}
↳ Non-Negative definite
↳ Correlation
↳ feature extraction
↳ feature matching.
↳ random variable
↳ Distribution

Probability:

↳ Experiment, sample space.

→ Complement

→ intersection $(A \cap B)$ → common elements to A & B.

★ → collectively exhaustive

→ mutually exclusive / disjoint ↗ $A \cap B = \phi$

Permutation & Combinations:

$0! = 1$ (Assumption or true?)

Conditional probability: B occurring, when A is

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$

Independent event: knowing probability of one event doesnt change the probability of another event

$$P(A/B) = P(A)$$
$$P(B/A) = P(B)$$

Discrete Probability Distribution:

→ distribution with equal probabilities is called uniform

formula:

$\mu = \Sigma x f(x)$ (mean of any discrete distribution of a random variable)

Characteristics of expected value } self study
$E(a) \to$ constant

Variance of uniform distribution.

$$Var(x) = \sigma^2 = \Sigma x^2 f(x) - \mu^2$$

Binomial Distribution:

number x of successes in n Bernoulli trials.

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x} \rightarrow \text{random variable.}$$

↳ probability of success at each trial

(n) number of trials

$q = 1-p$ (failure probability)

x – no. of successes in n independent trials (fixed 'n' trials)

x = no. of success (i.e varying)

Negative Binomial Distribution.

X – number of trials required to produce k successes

multinomial → random variable keeps of varying

–ve binomial → fix # success

trials (varying)

$$b^* = \binom{x-1}{k-1} p^{k-1} q^{(x-1)-(k-1)} p$$

$$b^*(x; k, p) = \binom{x-1}{k-1} p^k q^{x-k}$$

7^{th} Aug'25 - Extra class

Learning is associated with pattern matching.

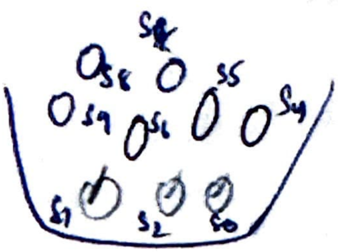
input — feature space

$$\{s_1, s_2, \dots, s_n\} \hookrightarrow \{(\mathbb{R}^m)_{s_1}, (\mathbb{R}^m)_{s_2}, \dots (\mathbb{R}^m)_{s_n}\}$$

output : Similar objects will be in same group.

$$\mathcal{D}(S_1, S_2) = T_1$$
$$\mathcal{D}(S_1, S_3) = T_2 \quad < T \Rightarrow S_1, S_2 \text{ are similar.}$$
$$\mathcal{D}(S_2, S_3) = T_3$$

T : threshold



$\Longrightarrow$ output

Ø what is this ? cherry

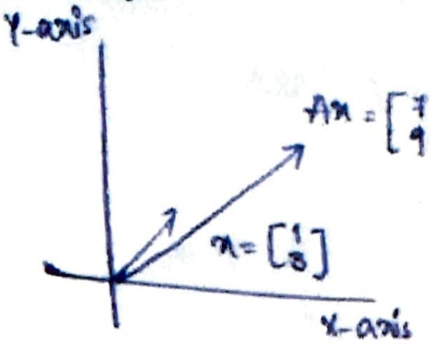
L_1 : Ø : cherry

L_2 : O - banana

L_3 : O - grape

$$\{(S_1, L_1)\ (S_2, L_1), \dots (S_9, L_3)\}$$
$$\downarrow$$
$$((\mathbb{R}^m)_{s_1}, L_1)\}$$

20th Aug'25 (Extra class)



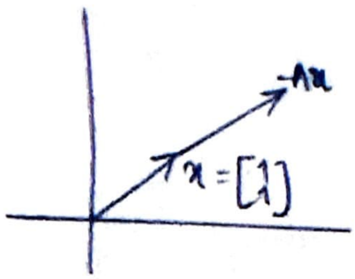
what happens when a matrix A hits a vector x?

↳ scalar

↳ scaled up & scaled down

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

give a specific time down vector



Eigen vector

$$Ax = \lambda x$$

eigen value

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

$$Ax = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Let, assume on day 0, k_1 students eat chinese food, k_2 students eat Indian food.

Chinese	Indian
k_1	k_2

$$V_0 = \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

→ on each subsequent day i, a fraction p of students ate chinese on Day i-1 continue to eat chinese on Day i and (1-p) shift to Indian food

$$V_1 = \begin{bmatrix} pk_1 + (1-q)k_2 \\ (1-p)k_1 + qk_2 \end{bmatrix}$$

⊙ ....

q Student ate Indian on Day i-1

(1-q) . ... Chinese

$$= \begin{bmatrix} p & 1-q \\ 1-p & q \end{bmatrix} \begin{bmatrix} k_1 \\ k_2 \end{bmatrix}$$

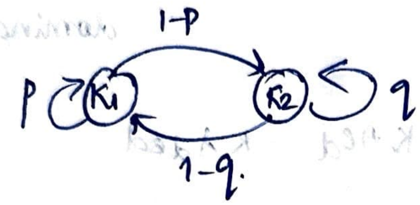
$$= MV_0$$

$$V_2 = M^2 V_0$$

$$\vdots$$

$$V_n = M^n V_0$$

everyday fractions:



Definition: let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of an $n \times n$ matrix A. λ_1 is called the <u>DOMINANT</u> eigen value

if $\lambda_1 \geq \lambda_i, \ \forall \ i: 2 \to n$

Def^n: A matrix M is called <u>stochastic Matrix</u>, if all the entire are positive, and sum of the elements in each elements is equal to 1.

Theorem: The largest (dominant) eigenvalue of ~~a~~ a stochastic matrix is <u>1</u>

<u>Theorem:</u> If A is a $n \times n$ square matrix with a dominant eigen value, then the sequence of vectors given by

$AV_0, A^2V_0, \ldots, A^nV_0, \ldots$ approaches a multiple of dominant eigen value of A

let e_d be the dominant eigen vector of M & $\lambda_d = 1$ the corresponding dominant eigen vector. the sequence

$MV_0, M^2V_0, M^3V_0, \ldots$

The exists an n s.t. $V_n = M^n V_0 = ke_d$.

what will happen for (n+1)?

$$V_{n+1} = MV_n = M(Ke_d) = K(Me_d) = K(\lambda_1 e_d)$$
$$= Ke_d$$

now, instead of stochastic matrix lets consider any square matrix A
let, p be time step at which the sequence

$x_0, Ax_0, A^2x_0, \ldots$ approaches to multiple of e_d

dominant eigen vector of A

$$A^p x_0 = Ke_d$$

$$A^{p+1} x_0 = A(A^p x_0) = KAe_d = K\lambda_d e_d$$

$$A^{p+2} x_0 = A(A^{p+1} x_0) = K\lambda_d A e_d = K\lambda_d^2 e_d$$

$$A^{p+n} x_0 = K\lambda_d^n e_d$$

In general if λ_d is dominant eigen value of A.

$|\lambda_d| > 1$ (explode), $\lambda_d = 1$ (steady)

$|\lambda_d| < 1$ ( )

Basics of Linear Algebra:

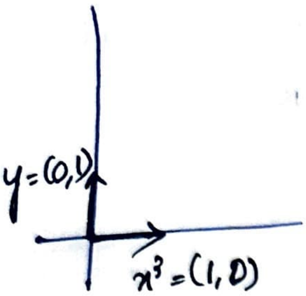
A set of vectors $\in \mathbb{R}^n$ is called a basis, if they are linearly independent and every vector $\in \mathbb{R}^n$ can be expressed as a linear continuation of these vectors

Linearly independent vectors: A set of n vectors $v_1, v_2, \ldots, v_n$ in linearly independent if no vector in the set can be expressed as a linear combination of remaining (n-1) vectors

In other words

$c_1v_1 + c_2v_2 + \ldots + c_nv_n = 0$ this eqn's only soln

(c_i's are scalar) is $c_1 = c_2 \ldots = c_n = 0$



Consider the space $\mathbb{R}^2$

vector $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Any vector $\begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2$, can be expressed as linear combination of these 2 vectors

$$\begin{bmatrix} a \\ b \end{bmatrix} = a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

further x & y are linearly independent

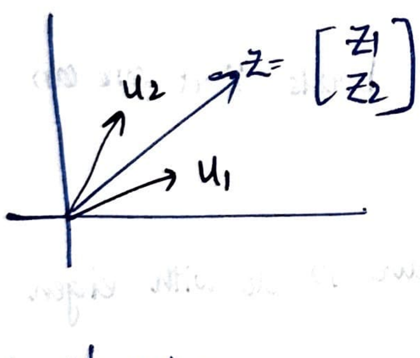
the only solution to $c_1 x + c_2 y = 0$ is $c_1 = c_2 = 0$

we could have chosen any 2 linearly independent vectors $\in \mathbb{R}^2$ as the basis

eg: $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ $\begin{bmatrix} 5 \\ 7 \end{bmatrix}$;

$$\begin{bmatrix} a \\ b \end{bmatrix} = x_1\begin{bmatrix} 2 \\ 3 \end{bmatrix} + x_2\begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

20th Aug'25 (2nd class)



In general, given a set of linear independent vectors $u_1, u_2, \dots, u_n \in \mathbb{R}^n$, we can express any vector $z \in \mathbb{R}^n$, as linear combination of these vectors

$$z = \alpha_1 \underline{u_1} + \alpha_2 \underline{u_2} + \dots + \alpha_n \underline{u_n}.$$

find α_i's using guassian elimination method

$$\alpha_1 \begin{bmatrix} u_{11} \\ u_{12} \\ \vdots \\ u_{1n} \end{bmatrix} + \alpha_2 \begin{bmatrix} u_{21} \\ u_{22} \\ \vdots \\ u_{2n} \end{bmatrix} + \dots + \alpha_n \begin{bmatrix} u_{n1} \\ u_{n2} \\ \vdots \\ u_{nn} \end{bmatrix}$$

$O(n^3) \to$ time complexity

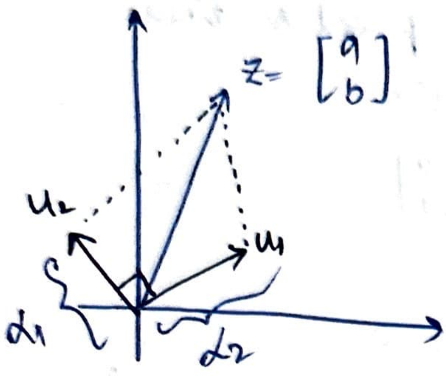
$$= \begin{bmatrix} u_{11} & \dots & u_{n1} \\ u_{12} & \dots & u_{n2} \\ \vdots & & \vdots \\ u_{1n} & & u_{nn} \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}$$

Orthogonal - perpendicular to each other

Orthonormal basis : Lets see if we have orthonormal basis

$u_i^T u_j = 0 \quad \forall \, i \neq j$

$u_i^T u_j = \|u_i\|^2 = 1$



$\alpha_1 = |\vec{z}| \cos\theta = |\vec{z}| \dfrac{z^T u_1}{|\vec{z}| \, |u_1|}$

$= z^T u_1$

Similarly

$\alpha_2 = z^T u_2$

we have

$z = \alpha_1 u_1 + \alpha_2 u_2 + \cdots \alpha_n u_n$

$u_1^T z = \alpha_1 u_1^T u_1 + \cdots + \alpha_1 u_1^T u_n = \alpha_1$.

we can directly find each α_1 using a ~~distant~~ dot product z & u_1 } $O(n)$ time complexity

for all α_1 : $O(n^2)$

→ An orthonormal basis is the most convenient basis that one can think of !!

Q) what does any of this orthogonal basis have to do with eigen vectors ?

— A) Turns out that the eigenvectors & can form a basis

- Eigenvectors of a square symmetric matrix even more special
- Thus they can form a very convenient basis.

why we would want to use eigenvectors as a basis instead of more normal coordinate axis?

Theorem: The eigenvectors of a matrix $A \in \mathbb{R}^{n \times n}$ having ~~differ~~ distinct eigenvalues are linearly independent

Theorem: The eigenvectors of a square symmetric matrix are orthogonal

hw: proof.

Eigen Value Decomposition:

→ let $u_1, u_2, \ldots, u_n$ be the eigenvector of a matrix A & $\lambda_1, \lambda_2, \ldots, \lambda_n$ the corresponding eigenvalues

→ consider matrix U where columns are $u_1, u_2, \ldots, u_n$

→ $$AU = A \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} = \begin{bmatrix} Au_1 & Au_2 & \cdots & Au_n \end{bmatrix} = \begin{bmatrix} \lambda_1 u_1 & \lambda_2 u_2 & \cdots & \lambda_n u_n \end{bmatrix}$$

(multiplying with A)

$$U\Lambda = \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & 0 & 0 \\ 0 & 0 & \lambda_2 & 0 & 0 \\ & & \vdots & \\ 0 & 0 & 0 & \lambda_n \end{bmatrix}$$

capital lambda

principal diagonal (eigen value)

$$\boxed{AU = U\lambda \neq \lambda U}$$

Λ is a diagonal matrix other depend ~~w~~ elements one are the eigen values of A

$$\boxed{\Lambda U = U\Lambda}$$

If U^{-1} exists, then we can write

$A = U\Lambda U^{-1}$ (eigenvalue decomposition)

$U^{-1}AU = \Lambda$ (diagonalization of Λ)

under what condition, U^{-1} exists?

if exists of U are linearly independent

— if A has n linearly independent eigenvectors

— A has n distinct eigen values.

sufficient condition.

if it is symmetric? (A)

the situation is more convenient and

eigenvectors are ortho~~gonal~~ normal

can be normalized by $u_i^T u_i = 1$

$$Q = U^T U = \begin{bmatrix} — u_1 — \\ — u_2 — \\ \vdots \\ — u_n — \end{bmatrix} \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_n \\ | & | & & | \end{bmatrix}$$

here,

each cell of $Q_{ij} = u_i^T u_j = 0$, if $i \neq j$

$= 1$, if $i = j$

$\boxed{U^T U = I}$ (identity matrix)

U^T is the inverse of U.

Theorem: given eigen value decomposition (EVD)

$$A = U \Sigma U^T$$

sequence $x_0, Ax_0, A^2x_0, \dots$ in terms of eigen value of A

n^{th} term will be?

↳ $U \Sigma^n U^T x_0$

Theorem: if A is a square symmetric matrix ($n \times n$) then the solution to the following optimization problem is given by the eigen vector corresponding to the largest eigen value of A

eqn:
$$\max\ x^T A x \quad \text{s.t.}\ \|x\| = 1$$

is given by the eigen vector corresponding to the largest eigen value of A.

(or)

$$\min\ x^T A x \quad \text{s.t.}\ \|x\| = 1$$

is given by the eigenvector corresponding to the smallest eigen value of A.

Sometimes you get $\min\ x^T A x$ s.t $\|x\| = 1$

sometimes, you get smallest eigen value of A

We are given a large no. of images of human faces (each image is of size 100×100

→ we like to represent and store the images using much fewer dimension $(n = 50 - 200)$

To construct a matrix $X \in \mathbb{R}^{m \times 10k}$

$$\hat{X} \in (\mathbb{R}^{m \times 200})$$

→ we retain top 200 dimensions corresponding to the top 200 eigen vectors of $X^T X \in \mathbb{R}^{n \times n}$

→ we can ~~conv~~ convert each eigen vector into a 200×200 matrix & treat this as image.

→ Let $v_1, v_2, \dots v_n$ be the eigen vectors of A and let $\lambda_1, \lambda_2, \dots, \lambda_n$ be the corresponding eigen value

$$AV_1 = \lambda_1 V_1, \dots, AV_n = \lambda_n V_n$$

if a vector $x \in \mathbb{R}^n$ is represented using $v_1, v_2, \dots v_n$ as basis them

$$x = \sum_{i=1}^{n} \alpha_i v_i, \quad Ax = \sum_{i=1}^{n} \alpha_i A v_i$$

$$= \sum_{i=1}^{n} \alpha_i \lambda_i v_i$$

→ The matrix multiplication reduces to scalar multiplication, if the eigen vectors of A are used as a basis.

→ so far, all the ~~dimens~~ discussion was centered on SQUARE MATRIX

→ what about rectangular matrix $A \in \mathbb{R}^{m \times n}$?

can we have eigen vectors?

→ It is possible to have $A_{m\times n}\, x_{n\times 1} = x_{n\times 1}$?

→ The result of $A_{m\times n}\, x_{n\times 1} \in \mathbb{R}^m$ ($x \in \mathbb{R}^n$)

→ So, do we miss out the advantage that a basis of eigen vector provides for square matrix?

(Matrix multiplication → Scalar multiplication)

Note: matrix $A_{m\times n}$ provides a transformation

$$\mathbb{R}^n \longrightarrow \mathbb{R}^m$$

* What if we could have pairs of vectors (v_1, u_1) $(v_2, u_2), \ldots, (v_k, u_k)$ s.t $v_i \in \mathbb{R}^n$, $u_i \in \mathbb{R}^m$ and

$$Av_i = s_i u_i$$

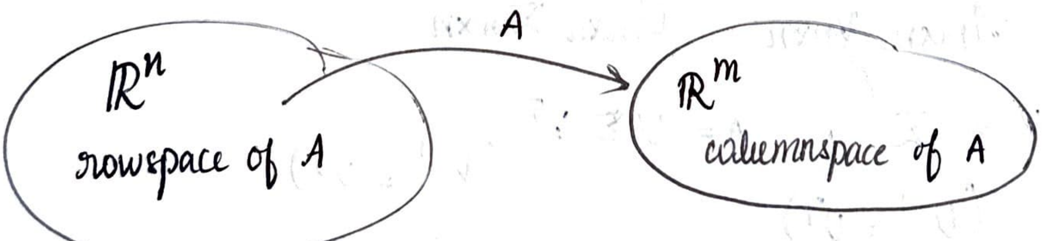
Assume, v_i's are orthogonal & may form basis $V \in \mathbb{R}^n$

$$i : i \longrightarrow n$$

Assume, v_i's $\longrightarrow 1:m$

* if every vector $x \in \mathbb{R}^n$ is represented using basis V

$$x = \sum_{i=1}^{k} \alpha_i v_i > Ax = \sum_{i=1}^{k} \alpha_i A v_i = \sum_{i=1}^{n} \alpha_i s_i u_i$$



dim = k = rank(A) dim = k = rank(A)

$\mathbb{R}^n$: space of all vector while can multiply with A to give Ax (this is the space of i/ps)

$\mathbb{R}^m$: space of all vectors which are outputs of f: Ax.

find U, V ← basis of O/p

basis of i/p

→ What do you mean by saying dim of rowspace in k?
if $x \in \mathbb{R}^n$ then why the dim is not n?

- It means that all possible vectors in $\mathbb{R}^n$ only a subspace of vectors lying in $\mathbb{R}^k$ can act as i/ps to Ax and produce non-zero o/p. The remaining vectors $\mathbb{R}^{n-k}$ will produce a zero o/p.

* we need only k dimension to represent x.

$$x = \sum_{i=1}^{k} \alpha_i v_i$$

$$Av_1 = \sigma_1 u_1, \quad Av_2 = \sigma_2 u_2, \quad \ldots \quad Av_k = \sigma_k u_k$$

$$A_{m\times n} V_{n\times k} = U_{m\times k} \Sigma_{k\times k}$$

↑ diagonal matrix

if we have k orthogonal vectors $V_{n\times k}$ then using Gram Schmidt orthogonalization, we can find $(n-k)$ more orthogonal vectors to compute the basis for $\mathbb{R}^n$ (similarly for U)

$$A_{m\times n} V_{n\times n} = U_{m\times n} \Sigma_{m\times n}$$

$$U^T A V = \Sigma \; ; \quad A = U \Sigma V^T$$

$(U^{-1} = U^T)$ $\quad (V^{-1} = V^T)$

Σ is a diagonal matrix with only the first k diagonal elements as non-zero.

→ How do we find V, U, Σ? (Think!)

Suppose V, U, Σ exists, then

$$A^T A = (U \Sigma V^T)^T (U \Sigma V^T)$$
$$= V \Sigma^T U^T U \Sigma V^T$$

eigen vector

$$AA^T = U \Sigma \Sigma^T U^T$$

eigen vector

Thus, U & V are the eigen vectors of AA^T and A^TA respectively and $\Sigma^2 = \Lambda$, where Λ is the diagonal matrix containing eigen values of A^TA

$$\begin{bmatrix} A \end{bmatrix}_{m\times n} = \begin{bmatrix} \vdots & \vdots & & | \\ U_1 & U_2 & \cdots & U_k \\ \vdots & \vdots & & | \end{bmatrix}_{m\times k} \begin{bmatrix} \delta_1 & 0 & 0 & 0 & 0 \\ 0 & \delta_2 & & & \\ 0 & & \ddots & & \\ 0 & & & & \delta_k \end{bmatrix} \begin{bmatrix} - V_1 - \\ - V_2 - \\ \vdots \\ - V_k - \end{bmatrix}_{k\times n}$$

Theorem:

$\sigma_i U_i V_i^T$ is the best rank-1 approximation of matrix A

$\sum_{i=1}^{k} \sigma_i V_i^T U_i$ is the best rank-k approximation of A.

Solution of $\min \|A - \hat{A}\|_F^2$ is given by

$\hat{A} = U_{:,k} \Sigma_{k*k} V_k^T$. ← all column.

[Minimizes the reconstruction error of A)

$\delta_i = \sqrt{\lambda_i} =$ singular matrix of A

$U =$ left singular matrix of A

$V =$ right singular matrix of A.